

KASHMIRA PATIL

Data & ML Engineer / Real-Time Systems / Graph ML / FastAPI

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PROFESSIONAL OVERVIEW

Engineer specializing in building production-grade, real-time data systems that transform complex datasets into actionable intelligence. Strong focus on system design, scalability, and performance optimization across machine learning and data engineering domains.

QFORCE — REAL-TIME GRAPH ML SYSTEM

Problem

Traditional stock prediction models fail to capture interdependencies between companies, limiting predictive performance.

Solution

Designed a graph-based system modeling relationships between companies using dynamic graph structures and deployed real-time inference APIs.

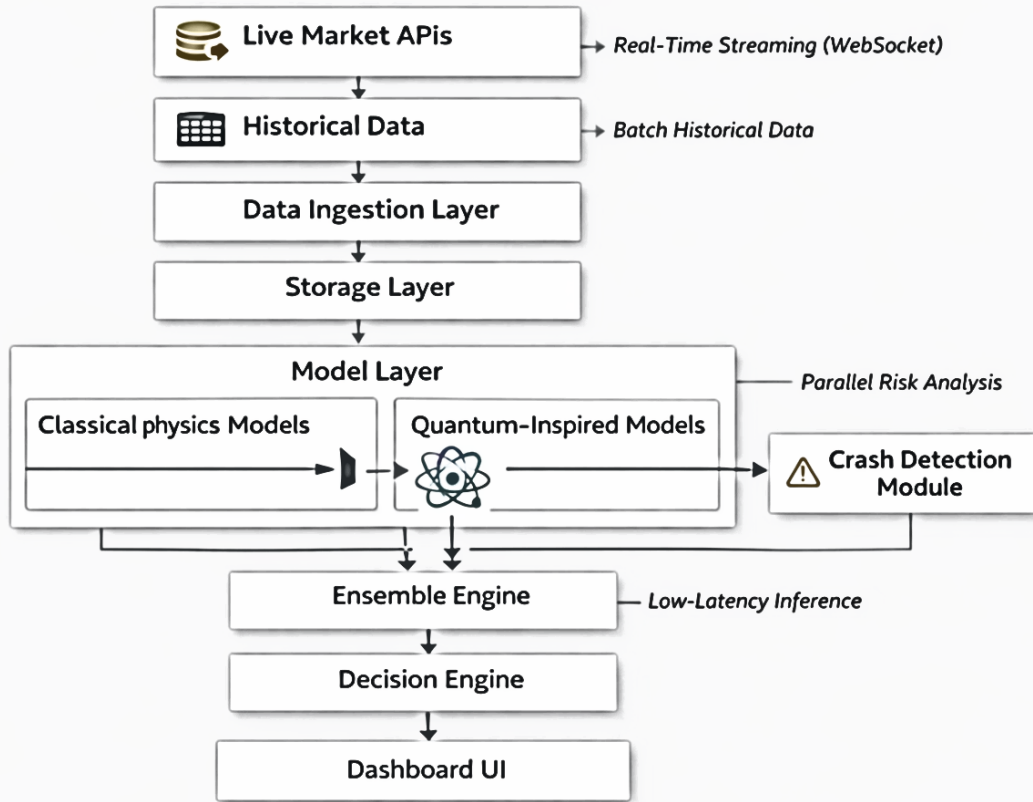
Architecture

- Data Ingestion: S&P 500 dataset (2018–2025)
- Feature Engineering: Graph construction & normalization
- Model Layer: Graph Neural Network
- Serving Layer: FastAPI real-time inference

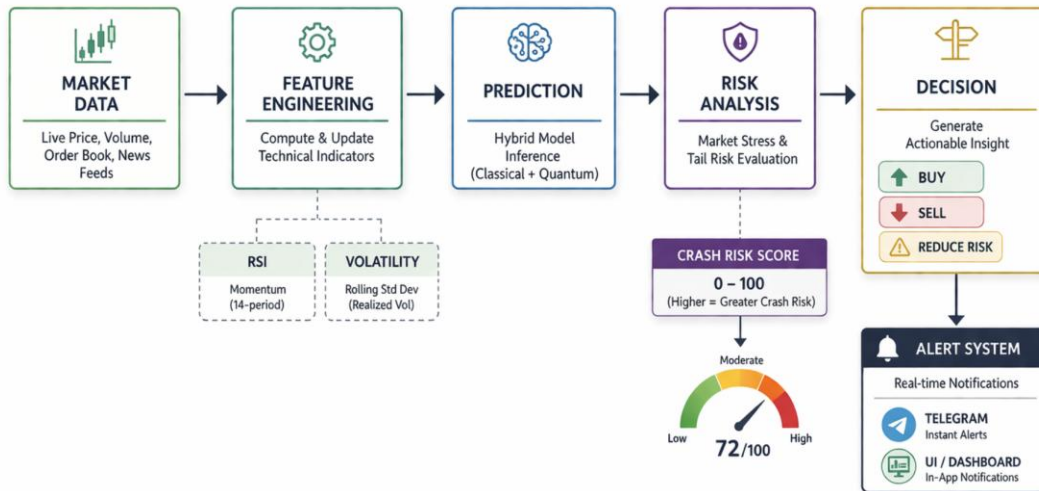
Impact

- Prediction error reduced: 6.1% → 4.9% MAPE
- Crash detection recall: 82%
- Inference latency: <2 seconds

System Architecture



Data Flow & Decision Pipeline



BOARDSIGHT — DECISION INTELLIGENCE SYSTEM

Problem

Lack of structured decision traceability in meetings leads to weak accountability and governance gaps.

System Design

Multi-layer AI system converting unstructured video/audio into structured decision analytics.

Core Architecture

- Perception Layer: Speaker detection and tracking
- Decision Detection: Keyword and context analysis
- Workflow Engine: Approval and escalation mapping
- Trace Layer: Structured reporting with audit integrity

Key Features

- Speaker dominance detection (~90% accuracy)
- Decision timestamp mapping
- Workflow trace generation
- SHA-256 based tamper-proof logs

Performance

- ≤1 sec/frame processing
- ~30 sec full trace generation
- Structured outputs (JSON/PDF)

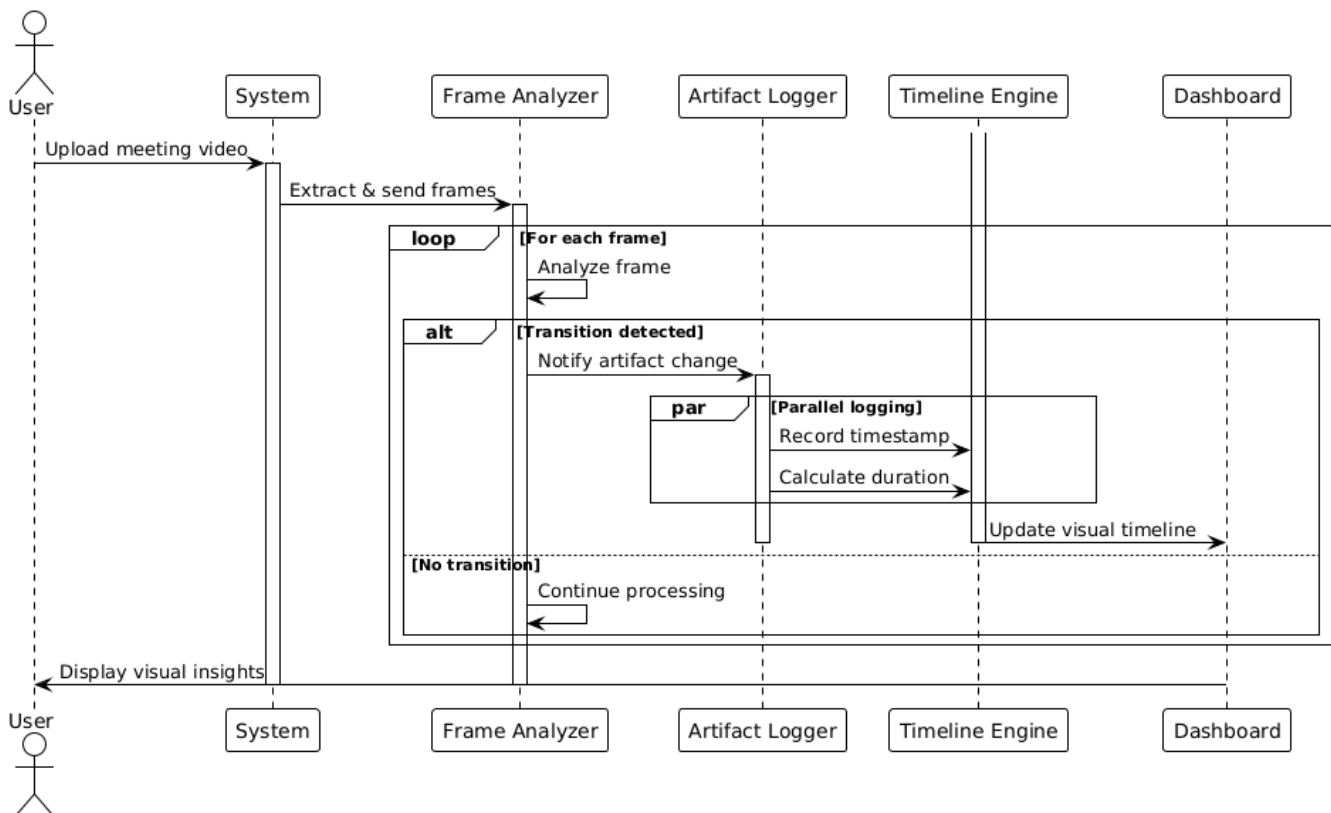


Fig: Sequence Diagram

BoardSight Meeting Activity Diagram

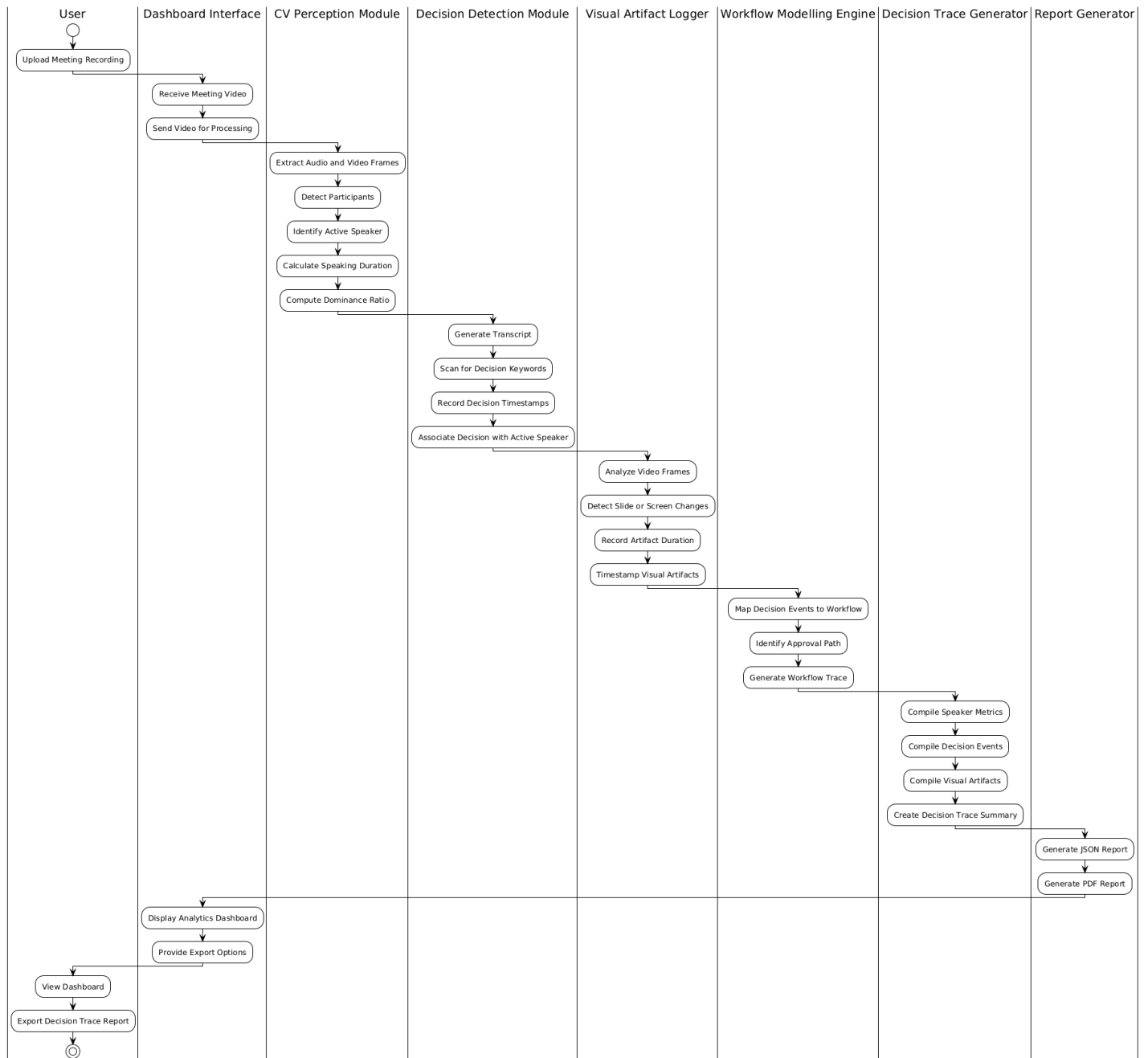


Fig: Activity Diagram

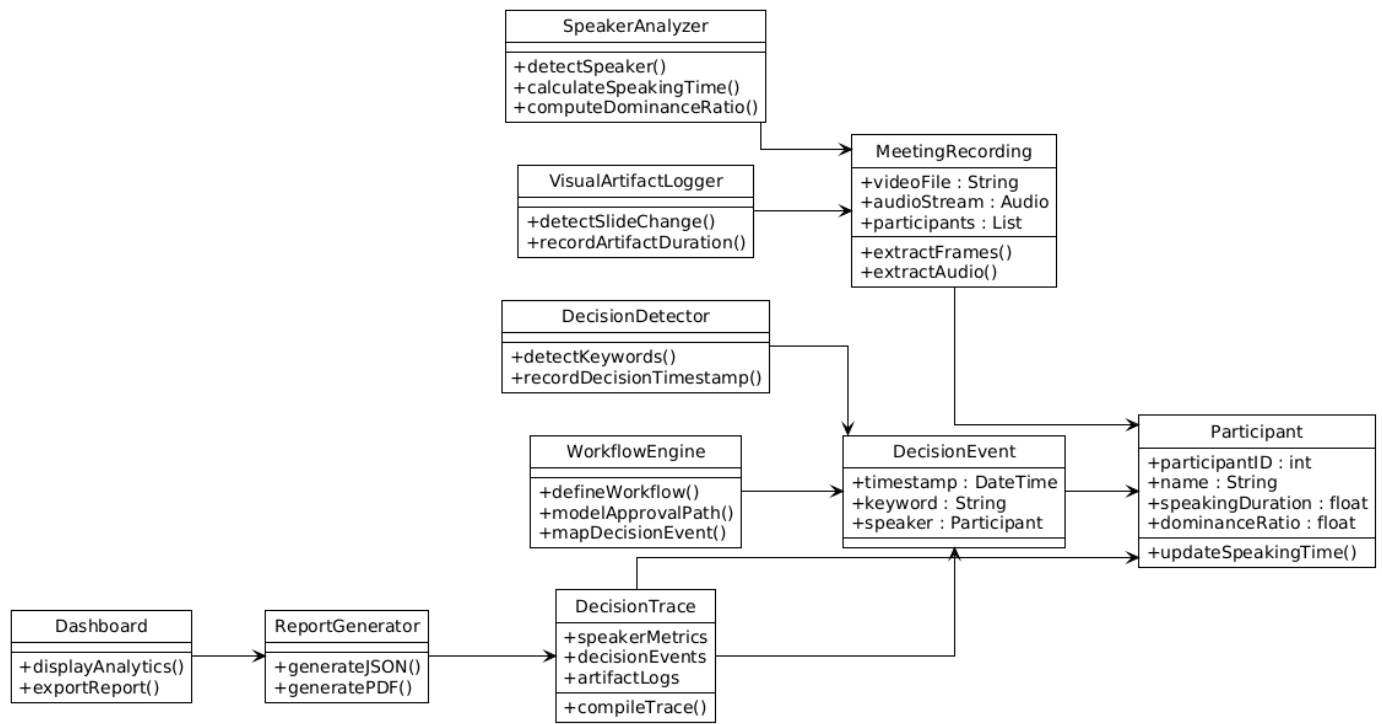


Fig: Class Diagram

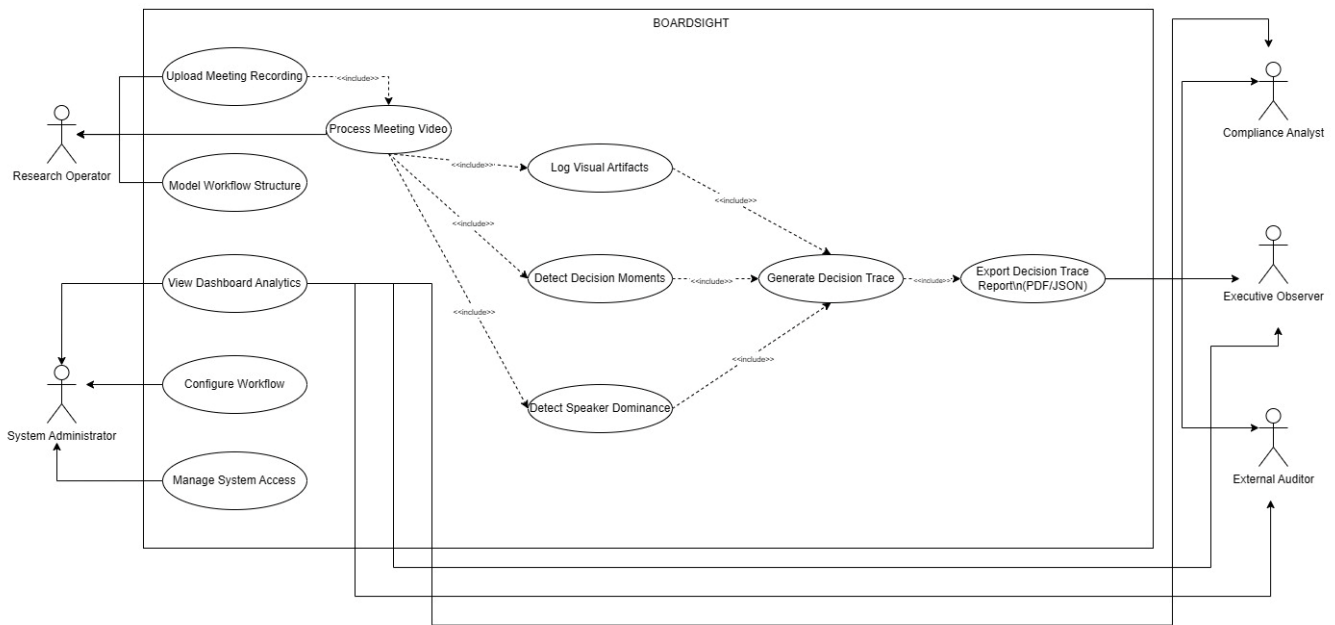


Fig: Use-Case Diagram

DAMIEN OS — INTELLIGENT AUTOMATION

Problem

Traditional OS interactions are manual, fragmented, and inefficient, lacking a unified interface to translate human intent into executable workflows.

Solution Overview

Designed a multi-modal intelligent system that converts natural language (voice/text) into structured, executable system-level actions using LLM-based intent parsing and modular automation pipelines.

System Architecture:

- **Input Layer:** Captures voice (ASR) and text inputs
- **Processing Layer:** Uses Gemini API to parse intent and generate structured JSON commands
- **Execution Layer:** Safely executes actions via Python automation (pyautogui, pywin32)
- **Data Layer:** Logs interactions, outcomes, and metadata in SQLite
- **Interface Layer:** React Application

System Pipeline

User Input → Intent Parsing (LLM) → Command Structuring → Execution Engine → Logging & Feedback

Key Contributions

- Designed **end-to-end orchestration pipeline** for converting unstructured input into structured executable workflows
- Implemented **LLM-based intent recognition system** with structured JSON outputs for reliable task mapping
- Built **modular execution engine** enabling OS-level automation (file operations, browser actions, Excel interactions)
- Developed **real-time interaction interface** using Streamlit + Flask for seamless user experience
- Integrated **speech recognition + text-to-speech** for full duplex voice interaction
- Engineered **audit logging system** to track commands, execution status, and timestamps for traceability
- Implemented **safety layer with confirmation mechanisms** for high-risk actions (preventing unintended execution)

System Design Thinking

- Transforms **unstructured human intent** → **structured machine-executable commands**
- Emphasizes **safety, auditability, and controlled automation**
- Modular architecture allows independent scaling of:
 - NLP layer
 - Execution layer
 - Interface layer

Performance & Capabilities

- Near real-time response for command execution
- Supports multi-modal interaction (voice + text)
- Handles diverse automation tasks:
 - Application control
 - File manipulation
 - Web actions
 - Excel operations
- Maintains **complete execution logs for transparency and debugging**

Advanced Features

- **Manual confirmation system** for sensitive operations
- **Kill-switch mechanism** to prevent runaway automation
- **Analytics dashboard** showing:
 - Intent distribution
 - Success rates
 - Usage trends
- **Structured logging via SQLite** enabling audit trails

Impact

- Reduced manual task execution effort through automation
- Enabled intuitive human-computer interaction via natural language
- Improved system usability through conversational interface + analytics

INDUSTRY EXPERIENCE

SDE Intern — Kash Precisions

- Automated cheque processing system (80% faster)
- Built FastAPI + MySQL backend
- Optimized async workflows

Data Science Intern — ZIDIO

- Built fraud detection pipeline (F1: 0.89)
 - Designed scalable preprocessing
 - Enabled real-time analytics
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CORE STRENGTHS

- Real-Time Data Systems
- Scalable ML Deployment
- Data Pipelines & Transformation
- System Design Thinking
- Workflow Automation